NON-EMPTINESS OF THE REGULAR PART OF THE AFFINE-QUIVER MODULI SPACE IS EQUIVALENT TO TWO SETS OF DOMINANCE INEQUALITIES

HIRAKU NAKAJIMA

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Allowed standard prerequisites (editorial).

- The Uhlenbeck partial compactifications X_zeta^circ, X_zeta^bullet and X_0 = C^2/Gamma and the moduli spaces M_zeta^circ, M_zeta^bullet, M_0 attached to them. Set up in the paragraph preceding the counted statement (source lines 2240-2261). Fixes the varieties whose regular parts the statement describes, and the notation lsp{t}{delta} w = 1 or >= 2 that splits the two numbered parts.
- The Weyl group of the affine root system, the elements alpha^c_h and e_i, and the weights of the irreducible integrable highest weight module V(w). Used in the statement at source lines 2269-2278 and inside the proof; the author's earlier Theorem at source lines 1219-1243 (label thm:CB) is the criterion the proof applies. Supplies the objects appearing in the two sets of inequalities and the alternative description by weights of V(w) in part (2).

Source reference. Original theorem 2.11: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1219–1219. Navigation pointer only: the counted proof invokes the author's earlier criterion by name for both directions, and the two formulations it needs are declared as named standard prerequisites, so the theorem's own statement and proof are not reproduced here.

Source reference. Original proposition 2.19: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1763–1763. Navigation pointer only: the counted proof uses the author's companion proposition on the strata of

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the same moduli space in one step, and its content is declared as a named standard prerequisite rather than reproduced.

Proposition 0.1 (cf. [6, 10.5, 10.8])**.** (1) *Suppose* ${}^t\delta\mathbf{w} = 1$. *Then* $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \neq \varnothing$ *if and only if* $\mathbf{w}-\mathbf{v}$ *is in the Weyl group orbit of* $\mathbf{w}$ *and satisfies*

$$ {}^t(\delta - \alpha_h^c)(\mathbf{w}-\mathbf{C}\mathbf{v}) \geq 0 \quad \text{for } c \in \mathcal{C}, \qquad {}^t\mathbf{e}_i(\mathbf{w}-\mathbf{C}\mathbf{v}) \geq 0 \quad \text{for } i \in I_0^0. $$

(2) *Suppose* ${}^t\delta\mathbf{w} \geq 2$. *Then* $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \neq \varnothing$ *if and only if* $\mathbf{w}-\mathbf{v}$ *is a weight of the irreducible integrable highest weight module* $V(\mathbf{w})$ *and the above inequalities hold.*

Proof. One can give a proof along arguments in [6, 10.5, 10.8], but we use Theorem 2.11 here.

If $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \neq \varnothing$, then $\mathbf{w}-\mathbf{v}$ is a weight of the irreducible integrable highest weight module by Theorem 2.11(2). Moreover two inequalities hold, as we have explained why $\widehat{\mathbf{w}}$ has nonnegative entries. If ${}^t\delta\mathbf{w} = 1$, the second proof of Proposition 2.19 shows $\mathbf{w}-\mathbf{v} \in W\cdot\mathbf{w}$ is also necessary.

For the converse, we check the criterion in Theorem 2.11. From our choice of $\zeta^\bullet$, a positive root $\beta^{(t)}$ with $\zeta^\bullet\cdot\beta^{(t)} = 0$ is one of the following:

(a) $m\delta$ for $m > 0$,
(b) $m\delta + \alpha$ for $m \geq 0$, $\alpha \in \Delta(I_0^0)_+$, or
(c) $m\delta - \alpha$ for $m > 0$, $\alpha \in \Delta(I_0^0)_+$,

where $\Delta(I_0^0)_+$ is the set of positive roots of the subroot system I_0^0. We have

$$ {}^t(m\delta)(\mathbf{w}-\mathbf{C}\mathbf{v}) = m\,{}^t\delta\mathbf{w} \geq 1, \qquad {}^t(m\delta+\alpha)(\mathbf{w}-\mathbf{C}\mathbf{v}) \geq 0, \qquad {}^t(m\delta-\alpha)(\mathbf{w}-\mathbf{C}\mathbf{v}) \geq 0 $$

from the assumption. Then

$$ \begin{aligned} &{}^t\mathbf{v}\left(\mathbf{w}-\tfrac{1}{2}\mathbf{C}\mathbf{v}\right) - {}^t\mathbf{v}^0\left(\mathbf{w}-\tfrac{1}{2}\mathbf{C}\mathbf{v}^0\right) - \sum p(\beta^{(t)}) \\ &= \sum_t {}^t\beta^{(t)}(\mathbf{w}-\mathbf{C}\mathbf{v}) - \#\{t \mid \beta^{(t)} \in \mathbb{Z}_{>0}\delta\} + \frac{1}{2}\left(\sum {}^t\beta^{(t)}\right)\mathbf{C}\left(\sum \beta^{(t)}\right) \end{aligned} $$

is nonnegative. Suppose that this is 0. Then we must have ${}^t\delta\mathbf{w} = 1$ and all $\beta^{(t)}$ are δ. This case is excluded in (1) from the assumption that $\mathbf{w}-\mathbf{v}$ is the Weyl group orbit of $\mathbf{w}$. (See the proof of Proposition 2.19.) □

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